

PREPARED FOR

Outkast Designs

TEST CA Z2

36.95882, -122.03836

Section 23, T11S, R2W, Mount Diablo Meridian

✓ CORS Elev Delta: 620 ft below nearest CORS station (P534, 13.1 mi) – Good baseline geometry.

COORDINATES	ELEVATION	STATE PLANE	FEMA ZONE	DECLINATION
36.9588°N 122.0384°W WGS84	52 ft NAVD88 · MSL	CA 3 (0403) · N 1,811,173.26 E 6,112,183.17 (US ft, NAD83)	Zone X 06087C0333F	12.6° E WMM 2026

APN **004-174-07** | ZONING **CN** | ACRES **0.108 ac**

BEST GNSS WINDOW **Thursday Jul 16 · 3:00 PM – 7:00 PM** | PDOP **0.97 · 33 sv** | NEAREST CORS
P534 · 13.1 mi ✓

REPORT CONTENTS

01	7-DAY GNSS FIELD RANKING	202	SITE & SIGNAL REPORT	3
03	BEST STATIC OBS WINDOW	404	EMERGENCY HOSPITALS	5
05	DAY-BY-DAY WEATHER + DRONE	606	AI FIELD ANALYSIS	13
07	WATER LEVELS & GAUGES	1408	FAULTS & SUBSIDENCE	15
09	STRUCTURAL PARAMETERS	1610	SOILS & ENGINEERING	16
11	MINES & MINERALS	1712	NATIVE PLANTS & WILDLIFE	18
13	FCC MICROWAVE PATHS	1914	SOLAR & PASSIVE DESIGN	21
15	AI SITE INTELLIGENCE BRIEF	2316	FIELD MAP · TOPO · DXF SITE PLAN	24



 [VIEW FIELD MAP](#) ➤

 [SITE SATELLITE](#) ➤

 [EMERGENCY MAP](#) ➤

SCAN FOR KMZ
Site file · Google Earth
↓ Download KMZ

RECORDER	ASSESSOR
Santa Cruz County Assessor · 701 Ocean St, Santa Cruz, CA 95060	Santa Cruz County Assessor · 701 Ocean St, Santa Cruz, CA 95060
	
TOPOSCOUT ★ A SITE INTELLIGENCE REPORT ★	
NEAREST CORP P534 · 13.1 mi	NEAREST BN G01954 · 0.2 mi
LAT 36.21° N / LONG 122.20° W	
GENERATED July 16, 2026	SYSTEM TopoScout · info@toposcout.com



INTERACTIVE REPORT

This report includes clickable links and QR codes throughout – tap or scan them to open maps, official data sources, and field tools directly. Your site is also available as a drafting-ready CAD (DXF) file with georeferenced aerials. Add it anytime at toposcout.com.

SITE & SIGNAL REPORT

COORDINATES

36.9588, -122.0384

WGS84 / NAD83

ELEVATION

52 ft

Above MSL (approx)

MAGNETIC DECLINATION

12.6° E

Compass reads 12.6° too far W

FEMA FLOOD ZONE

X

AREA OF MINIMAL FLOOD HAZARD · Panel 06087C0333F

IONOSPHERIC CONDITIONS

PEAK KP INDEX

4.7

AVG KP

2.3

SOLAR FLUX F10.7

138 sfu

TEC (VTEC) ▲ DEGRADED

20.5 TECU

-3.5 TECU vs median

ASSESSMENT

MODERATE ACTIVITY

Peak Kp 4.7 (avg 2.3). Solar flux F10.7 = 138 sfu. Moderate geomagnetic activity. Some ionospheric delay possible. Static sessions should use longer occupation times.

GNSS / DOP SUMMARY

DATE	AVG PDOP	BEST 4-HR WINDOW	AVG SATS	RATING
Thu, Jul 16	1.01	3:00 PM – 7:00 PM	33	Good
Fri, Jul 17	1.00	3:00 PM – 7:00 PM	33	Good
Sat, Jul 18	0.97	3:00 PM – 7:00 PM	35	Ideal
Sun, Jul 19	0.99	2:00 PM – 6:00 PM	35	Ideal
Mon, Jul 20	0.98	6:00 AM – 10:00 AM	35	Ideal
Tue, Jul 21	0.99	3:00 PM – 7:00 PM	33	Ideal
Wed, Jul 22	1.01	4:00 PM – 8:00 PM	34	Good

GPS + GLONASS + Galileo + BeiDou MEO · Full multi-constellation model · 15° elevation mask · Per-day independent computation
GNSS satellite geometry is accurate within a 50-mile radius of the entered location – orbital mechanics do not change meaningfully at ground-level distances. Weather data is specific to the entered coordinates.

Days analyzed: 7 Recommended work days: 5 of 7 Best PDOP: 0.97

►

RECOMMENDED WINDOW

Mon, Jul 20

6:00 AM – 10:00 AM

Avg wind 3 mph · Gusts 3 mph

Ideal conditions — 4-hr static occupation

ALTERNATE WINDOW

Tue, Jul 21

4:00 PM – 8:00 PM

Avg wind 5 mph · Gusts 5 mph

Ideal conditions — 4-hr static occupation

🔥

WILDFIRE PROXIMITY

No Active Fires Within 150 Miles

No active fires as of report date (July 16, 2026) · Conditions may change – verify before fieldwork

📶

GPS CONSTELLATION HEALTH

⚠️ 1 Satellite Currently Unhealthy – PRN 11

Active outage per NAVCEN · Affected satellites auto-excluded by receiver

🌿 Land Cover & Multipath Risk

SKY OBSTRUCTION

Minor Obstruction

Low vegetation or sparse development nearby — minor sky obstruction possible.

MULTIPATH RISK

Low Multipath

Minor reflective surfaces present — multipath generally manageable.

RECOMMENDED ELEVATION MASK

13° Minor vegetation — slight horizon reduction

✓ PDOP Check at 13°: 0.8–1.1 across survey windows

PDOP remains ≤ 1.1 at 13° — geometry confirmed acceptable

5-POINT SAMPLE (CENTER + N/S/E/W · ~100 M)

Point	Cover Type	Obstruction	Multipath
Center	Dwarf Scrub	Low	Low
North	Dwarf Scrub	Low	Low
South	Dwarf Scrub	Low	Low
East	Dwarf Scrub	Low	Low
West	Dwarf Scrub	Low	Low

Dominant cover: Dwarf Scrub · Source: USGS NLCD 2024 via MRLC · 30 m resolution · Industry standard mask guidance per NGS field procedures

High 82°F

Low 54°F

Wind max 15 mph

Gusts 17 mph

Precip 0.00"

Sunrise 6:01 AM

Sunset 8:26 PM

New Moon 3%

GNSS: Avg PDOP **1.01** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.97, 33 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.93		32 sv		IDEAL
7:00 AM	0.99		32 sv		IDEAL
8:00 AM	0.87		36 sv		IDEAL
9:00 AM	1.20		27 sv		GOOD
10:00 AM	1.11		27 sv		GOOD
11:00 AM	1.05		27 sv		GOOD
12:00 PM	0.90		31 sv		IDEAL
1:00 PM	0.99		30 sv		IDEAL
2:00 PM	1.13		29 sv		GOOD
3:00 PM	0.95		33 sv		IDEAL
4:00 PM	0.95		35 sv		IDEAL
5:00 PM	0.90		35 sv		IDEAL
6:00 PM	1.08		29 sv		GOOD
7:00 PM	1.13		27 sv		GOOD



GO – GOOD CONDITIONS

Clear sky | Wind: 14.6 mph (gusts 17 mph) | Precip: 0" | Temp: 54.3°–81.8°F



DRONE WIND ANALYSIS



GO Peak: 14.6/17 mph WNW

* Sunrise: 6:01 AM ☾ Sunset: 8:26 PM fly window clipped to daylight

✓ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

🌀 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	2/2 mph		S
6 AM	1/1 mph		SSE
7 AM	3/3 mph		ESE
8 AM	2/3 mph		SE
9 AM	4/4 mph		S
10 AM	3/3 mph		S
11 AM	3/3 mph		WSW
12 PM	7/8 mph		W
1 PM	10/11 mph		WNW
2 PM	13/16 mph		WNW
3 PM	15/17 mph		WNW
4 PM	14/17 mph		WNW
5 PM	13/16 mph		WNW
6 PM	12/15 mph		WNW
7 PM	10/14 mph		NW
8 PM	8/13 mph		WNW

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 73°F

Low 53°F

Wind max 14 mph

Gusts 17 mph

Precip 0.00"

Sunrise 6:02 AM

Sunset 8:26 PM

Waxing Crescent 8%

GNSS: Avg PDOP **1.00** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.93, 33 sats avg)

GNSS / DOP - HOURLY					
	PDOP		SATS		RATING
6:00 AM	0.91		34 sv		IDEAL
7:00 AM	1.26		28 sv		GOOD
8:00 AM	0.91		35 sv		IDEAL
9:00 AM	0.93		33 sv		IDEAL
10:00 AM	0.99		30 sv		IDEAL
11:00 AM	0.96		28 sv		IDEAL
12:00 PM	0.92		31 sv		IDEAL
1:00 PM	1.12		27 sv		GOOD
2:00 PM	1.04		31 sv		GOOD
3:00 PM	0.94		34 sv		IDEAL
4:00 PM	0.93		35 sv		IDEAL
5:00 PM	0.95		33 sv		IDEAL
6:00 PM	0.91		31 sv		IDEAL
7:00 PM	1.17		25 sv		GOOD



GO – GOOD CONDITIONS

Partly cloudy | Wind: 14.2 mph (gusts 16.6 mph) | Precip: 0" | Temp: 53°–72.5°F

 DRONE WIND ANALYSIS

 **GO** Peak: 14.2/16.6 mph W

* Sunrise: 6:02 AM ☾ Sunset: 8:26 PM fly window clipped to daylight

 Best flight window: 7:00 AM – 9:00 PM (14 hrs)

 Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	4/5 mph		NW
6 AM	2/3 mph		NNW
7 AM	2/3 mph		ENE
8 AM	2/2 mph		SE
9 AM	2/2 mph		SSW
10 AM	4/5 mph		WSW
11 AM	10/11 mph		W
12 PM	13/14 mph		W
1 PM	13/15 mph		W
2 PM	14/17 mph		W
3 PM	13/14 mph		W
4 PM	10/11 mph		WSW
5 PM	7/7 mph		WSW
6 PM	4/3 mph		SSW
7 PM	2/6 mph		ESE
8 PM	3/5 mph		E

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS  NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#3

Saturday, Jul 18

WEEKEND

Clear sky

8 mph

GO

High 72°F

Low 57°F

Wind max 8 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:02 AM

Sunset 8:25 PM

Waxing Crescent 15%

GNSS: Avg PDOP 0.97 · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.92, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.91	34 sv	IDEAL
7:00 AM	1.02	32 sv	GOOD
8:00 AM	1.03	32 sv	GOOD
9:00 AM	0.94	33 sv	IDEAL
10:00 AM	1.01	30 sv	GOOD
11:00 AM	1.00	29 sv	GOOD
12:00 PM	0.90	32 sv	IDEAL
1:00 PM	1.02	31 sv	GOOD
2:00 PM	0.94	34 sv	IDEAL
3:00 PM	0.88	36 sv	IDEAL
4:00 PM	0.93	36 sv	IDEAL
5:00 PM	0.99	32 sv	IDEAL
6:00 PM	0.89	35 sv	IDEAL
7:00 PM	1.11	30 sv	GOOD



GO - GOOD CONDITIONS

Clear sky | Wind: 8.2 mph (gusts 6.3 mph) | Precip: 0" | Temp: 56.5°–72.1°F



DRONE WIND ANALYSIS



GO Peak: 8.2/6.3 mph SSE

* Sunrise: 6:02 AM Sunset: 8:25 PM fly window clipped to daylight



Best flight window: 7:00 AM – 9:00 PM (14 hrs)



Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	4/3 mph		SE
6 AM	4/3 mph		SE
7 AM	3/3 mph		SE
8 AM	3/2 mph		SSE
9 AM	4/2 mph		S
10 AM	5/3 mph		SSW
11 AM	7/4 mph		SSW
12 PM	8/4 mph		SW
1 PM	8/5 mph		SW
2 PM	8/6 mph		SW
3 PM	7/6 mph		SW
4 PM	7/6 mph		SSW
5 PM	6/6 mph		SSW
6 PM	4/3 mph		SSW
7 PM	2/3 mph		SSE
8 PM	3/5 mph		ESE

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY

Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107

→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)

Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

#4

Sunday, Jul 19

WEEKEND

Overcast

9 mph

GO

High 70°F | Low 57°F | Wind max 9 mph | Gusts 8 mph | Precip 0.00" | Sunrise 6:03 AM | Sunset 8:25 PM

Waxing Crescent 23%

GNSS: Avg PDOP 0.99 · Best 4-hr window 2:00 PM – 6:00 PM (PDOP 0.94, 35 sats avg)

GNSS / DOP - HOURLY

	PDOP	SATS	RATING
6:00 AM	0.96	34 sv	IDEAL
7:00 AM	1.04	29 sv	GOOD
8:00 AM	0.95	32 sv	IDEAL
9:00 AM	0.91	32 sv	IDEAL
10:00 AM	1.13	27 sv	GOOD
11:00 AM	1.14	25 sv	GOOD
12:00 PM	1.03	27 sv	GOOD
1:00 PM	0.93	31 sv	IDEAL
2:00 PM	0.91	35 sv	IDEAL
3:00 PM	1.02	32 sv	GOOD
4:00 PM	0.90	37 sv	IDEAL
5:00 PM	0.92	34 sv	IDEAL
6:00 PM	0.98	32 sv	IDEAL
7:00 PM	1.07	29 sv	GOOD



GO - GOOD CONDITIONS

Partly cloudy | Wind: 8.7 mph (gusts 8.1 mph) | Precip: 0" | Temp: 57.4°-70.3°F



DRONE WIND ANALYSIS



GO Peak: 8.7/8.1 mph SSW

* Sunrise: 6:03 AM | Sunset: 8:25 PM | fly window clipped to daylight



Best flight window: 7:00 AM – 9:00 PM (14 hrs)



Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≠turbulence	DIR
5 AM	2/2 mph		SSE
6 AM	2/2 mph		SSE
7 AM	3/2 mph		SE
8 AM	5/3 mph		SSE
9 AM	4/2 mph		SSE
10 AM	4/2 mph		SSW
11 AM	7/3 mph		SSW
12 PM	8/4 mph		SW
1 PM	7/5 mph		SSW
2 PM	8/6 mph		SW
3 PM	9/8 mph		SW
4 PM	7/7 mph		SW
5 PM	7/7 mph		SW
6 PM	7/6 mph		SSW
7 PM	4/5 mph		SSW
8 PM	2/3 mph		SW

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 71°F

Low 59°F

Wind max 8 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:04 AM

Sunset 8:24 PM

Waxing Crescent 33%

GNSS: Avg PDOP **0.98** · Best 4-hr window 6:00 AM – 10:00 AM (PDOP 0.92, 35 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.88	35 sv	IDEAL
7:00 AM	0.97	34 sv	IDEAL
8:00 AM	0.88	37 sv	IDEAL
9:00 AM	0.95	32 sv	IDEAL
10:00 AM	0.94	29 sv	IDEAL
11:00 AM	1.11	25 sv	GOOD
12:00 PM	1.00	28 sv	GOOD
1:00 PM	1.02	27 sv	GOOD
2:00 PM	0.99	33 sv	IDEAL
3:00 PM	0.98	34 sv	IDEAL
4:00 PM	0.94	33 sv	IDEAL
5:00 PM	0.97	32 sv	IDEAL
6:00 PM	0.92	34 sv	IDEAL
7:00 PM	1.19	28 sv	GOOD

GO – GOOD CONDITIONS

Partly cloudy | Wind: 7.8 mph (gusts 6 mph) | Precip: 0" | Temp: 58.5°–71.1°F

DRONE WIND ANALYSIS ✓ GO Peak: 7.8/6 mph S

* Sunrise: 6:04 AM ☾ Sunset: 8:24 PM fly window clipped to daylight

☒ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≤turbulence	DIR
5 AM	3/2 mph		ESE
6 AM	4/3 mph		ESE
7 AM	2/2 mph		SE
8 AM	3/3 mph		SSW
9 AM	3/2 mph		SE
10 AM	8/5 mph		SW
11 AM	7/4 mph		SW
12 PM	7/5 mph		SW
1 PM	7/6 mph		SW
2 PM	7/6 mph		WSW
3 PM	6/6 mph		SW
4 PM	7/6 mph		SW
5 PM	6/5 mph		SSW
6 PM	5/4 mph		SSW
7 PM	3/3 mph		SSW
8 PM	2/2 mph		S

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ NO TFRs WITHIN 30 MI [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 77°F

Low 60°F

Wind max 8 mph

Gusts 5 mph

Precip 0.00"

Sunrise 6:05 AM

Sunset 8:24 PM

Waxing Crescent 43%

GNSS: Avg PDOP **0.99** · Best 4-hr window 3:00 PM – 7:00 PM (PDOP 0.94, 33 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	0.95	32 sv	IDEAL
7:00 AM	0.97	34 sv	IDEAL
8:00 AM	0.89	35 sv	IDEAL
9:00 AM	1.13	30 sv	GOOD
10:00 AM	1.01	31 sv	GOOD
11:00 AM	1.07	28 sv	GOOD
12:00 PM	0.96	30 sv	IDEAL
1:00 PM	1.06	28 sv	GOOD
2:00 PM	1.03	30 sv	GOOD
3:00 PM	0.94	33 sv	IDEAL
4:00 PM	0.94	36 sv	IDEAL
5:00 PM	0.96	32 sv	IDEAL
6:00 PM	0.93	31 sv	IDEAL
7:00 PM	1.03	29 sv	GOOD

GO — GOOD CONDITIONS
Partly cloudy | Wind: 7.5 mph (gusts 5.1 mph) | Precip: 0" | Temp: 60°–76.6°F

DRONE WIND ANALYSIS ✓ **GO** Peak: 7.5/5.1 mph SSE

* Sunrise: 6:05 AM ☾ Sunset: 8:24 PM fly window clipped to daylight

☒ Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	1/1 mph		SW
6 AM	2/2 mph		S
7 AM	4/3 mph		SSE
8 AM	5/3 mph		SSE
9 AM	5/3 mph		S
10 AM	5/3 mph		S
11 AM	5/3 mph		SSW
12 PM	6/4 mph		SSW
1 PM	7/4 mph		SW
2 PM	8/5 mph		SW
3 PM	7/4 mph		SSW
4 PM	5/4 mph		SSW
5 PM	5/4 mph		S
6 PM	4/4 mph		SSE
7 PM	4/5 mph		SSE
8 PM	4/5 mph		SE

BEST WINDOW
7:00 AM – 9:00 PM
Ideal conditions — calm winds, clean operations all day

AIRSPACE **CLASS A** ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS ☒ **NO TFRs WITHIN 30 MI** [Check B4UFLY](#) ↗
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

High 76°F

Low 60°F

Wind max 9 mph

Gusts 6 mph

Precip 0.00"

Sunrise 6:05 AM

Sunset 8:23 PM

First Quarter 53%

GNSS: Avg PDOP **1.01** · Best 4-hr window 4:00 PM – 8:00 PM (PDOP 0.91, 34 sats avg)

GNSS / DOP - HOURLY			
	PDOP	SATS	RATING
6:00 AM	1.15	29 sv	GOOD
7:00 AM	1.05	30 sv	GOOD
8:00 AM	0.98	34 sv	IDEAL
9:00 AM	1.05	30 sv	GOOD
10:00 AM	1.02	29 sv	GOOD
11:00 AM	1.17	25 sv	GOOD
12:00 PM	0.97	28 sv	IDEAL
1:00 PM	1.01	29 sv	GOOD
2:00 PM	0.98	34 sv	IDEAL
3:00 PM	1.17	33 sv	GOOD
4:00 PM	0.87	36 sv	IDEAL
5:00 PM	0.89	34 sv	IDEAL
6:00 PM	0.92	32 sv	IDEAL
7:00 PM	0.94	32 sv	IDEAL

GO — GOOD CONDITIONS

Partly cloudy | Wind: 8.6 mph (gusts 5.8 mph) | Precip: 0" | Temp: 60.1°–75.7°F

DRONE WIND ANALYSIS

GO Peak: 8.6/5.8 mph SSE

* Sunrise: 6:05 AM ☾ Sunset: 8:23 PM fly window clipped to daylight

Best flight window: 7:00 AM – 9:00 PM (14 hrs)

Wind direction shifting — plan mission headings carefully

	SUSTAINED / GUSTS mph	≈turbulence	DIR
5 AM	3/3 mph		ESE
6 AM	3/2 mph		ESE
7 AM	3/2 mph		SE
8 AM	3/2 mph		SSE
9 AM	3/3 mph		S
10 AM	5/3 mph		SSW
11 AM	7/4 mph		SSW
12 PM	8/5 mph		SW
1 PM	9/5 mph		SW
2 PM	8/6 mph		SW
3 PM	6/6 mph		SSW
4 PM	5/5 mph		SSE
5 PM	6/5 mph		SE
6 PM	7/6 mph		SE
7 PM	7/6 mph		ESE
8 PM	7/6 mph		ESE

BEST WINDOW

7:00 AM – 9:00 PM

Ideal conditions — calm winds, clean operations all day

AIRSPACE CLASS A ▶ FREE TO FLY
Class G uncontrolled airspace — fly under 400ft AGL, follow Part 107
→ TFR STATUS NO TFRs WITHIN 30 MI [Check B4UFLY ↗](#)
Current TFRs only · Always verify airspace day-of with B4UFLY or Aloft

FIELD PLANNING REPORT

JOB: SG-THZEJV | SITE: N/A | LOCATION: 36.9588, -122.0384 (SANTA CRUZ COUNTY)

REPORT GENERATED: Thursday, July 16, 2026

SURVEY WINDOW: July 16–22, 2026 (7 days)

=====

TODAY (THURSDAY, JULY 16) ASSESSMENT

DECISION: GO

RATIONALE:

Excellent PDOP (0.97 during 3:00 PM–7:00 PM window, 33 satellites). Clear skies, moderate temperatures (82°F high), light winds (15 mph max). Ionospheric activity moderate (Kp 4.7 peak) but manageable with standard static protocols. No precipitation. Site conditions favorable (dwarf scrub, low sky obstruction, 13° recommended mask). GPS PRN 11 unhealthy and PRN 26 maintenance window (00:35–12:35 UTC Friday) do not impact today's operations.

RECOMMENDATION: Deploy today if crew is available. Best window 3:00 PM–7:00 PM.

=====

MONDAY CALL (JULY 20 MOBILIZATION DECISION)

DECISION: GO

RATIONALE:

Monday ranks 3rd overall for the week. PDOP 0.92 during morning window (6:00 AM–10:00 AM, 35 satellites). Overcast conditions pose no survey impediment; zero precipitation forecasted. Ionospheric Kp declining trend by mid-week supports quality static sessions. Early morning window excellent for crew efficiency. No satellite health issues active by Monday.

RECOMMENDATION: Mobilize crews as planned. Prioritize 6:00 AM–10:00 AM slot for primary observations.

=====

WEEK RANKING (BEST TO WORST)

RANK 1: WEDNESDAY, JULY 22

PDOP 0.91 (best geometry all week), 34 sats, 4:00 PM–8:00 PM window.

Overcast, no precip, moderate winds. Ionospheric activity subsiding.

RANK 2: FRIDAY, JULY 17

PDOP 0.93, 33 sats, 3:00 PM–7:00 PM window.

Overcast (minimal impact), zero precip. PRN 26 maintenance 00:35–12:35 UTC (afternoon window unaffected).

RANK 3: MONDAY, JULY 20

PDOP 0.92, 35 sats, 6:00 AM–10:00 AM window.

Overcast, zero precip, light winds. Early morning advantage.

RANK 4: SATURDAY, JULY 18 [WEEKEND]

PDOP 0.92, 35 sats, 3:00 PM–7:00 PM window.

Clear skies, lightest winds (8 mph). Excellent conditions but weekend constraints.

RANK 5: THURSDAY, JULY 16 [TODAY]

PDOP 0.97, 33 sats, 3:00 PM–7:00 PM window.

Clear skies, immediate availability. Slightly elevated ionospheric activity.

RANK 6: SUNDAY, JULY 19 [WEEKEND]

PDOP 0.94, 35 sats, 2:00 PM–6:00 PM window.

Overcast, light winds. Weekend; ranks below weekday equivalents.

RANK 7: TUESDAY, JULY 21

PDOP 0.94, 33 sats, 3:00 PM–7:00 PM window.

Overcast, zero precip. Satellite count (33) and PDOP (0.94) slightly lower than competing days.

TOP 5 RECOMMENDED WORK CANDIDATES: Wed 7/22, Fri 7/17, Mon 7/20, Sat 7/18, Thu 7/16

=====

BEST STATIC WINDOW (4-HOUR CONSECUTIVE)

RECOMMENDATION: WEDNESDAY, JULY 22, 4:00 PM–8:00 PM (LOCAL TIME)

RATIONALE:

Best combined PDOP geometry (0.91), highest satellite count (34), stable ionosphere (Kp subsiding). Overcast does not degrade GNSS; zero precipitation risk. If weather delays force rescheduling, FRIDAY 3:00 PM–7:00 PM is secondary choice (PDOP 0.93, same window structure).

OCCUPY TIME: Recommend 4+ hours at primary points to mitigate residual ionospheric delay (TEC slightly below median). At least 2 static baselines recommended.

=====

DRONE ASSESSMENT (BY DAY)

Thursday 7/16: FLYABLE

Clear, max wind 15 mph, excellent visibility. Optimal for aerial surveys.

Friday 7/17: CAUTION

Overcast, max wind 14 mph. Flyable but reduced visibility; confirm airspace and battery performance.

Saturday 7/18: FLYABLE

Clear, min wind (8 mph), ideal conditions. Weekend constraints apply.

Sunday 7/19: CAUTION

Overcast, max wind 9 mph. Marginal visibility; ops possible with crew experience.

Monday 7/20: CAUTION

Overcast, max wind 8 mph. Flyable but reduced ceiling; early morning preferred before cloud buildup.

Tuesday 7/21: CAUTION

Overcast, max wind 8 mph. Similar to Monday; marginal conditions for extended ops.

Wednesday 7/22: FLYABLE

Overcast but stable, min wind (9 mph), expected light wind. Good operational window.

=====

FIELD WARNINGS

1. IONOSPHERIC DELAY — TEC anomaly (–3.5 TECU below 30-day median) flagged as degraded quality estimate. Increase static observation times by 1 hour minimum over baseline to achieve cm-level accuracy. Use dual-frequency receivers mandatory.

2. GPS PRN 11 UNUSABLE — Receiver firmware will auto-exclude. Nominal PDOP values already account for absence. No operational impact if multi-constellation receiver (GPS+GLONASS+Galileo+BeiDou). Verify receiver health before deployment.

3. PRN 26 MAINTENANCE FRIDAY 00:35–12:35 UTC — Does not affect afternoon/evening survey windows (15:00–19:00 local = 22:35 UTC–02:35 UTC+1). Informational; no action required.

4. SITE ELEVATION MASK 13° — Dwarf scrub terrain low-risk. Deploy antenna on tripod minimum 2 m above ground. Verify 13° horizon mask acceptance before occupation.

5. NO WILDFIRE THREAT — All-clear from NIFC. Santa Cruz County operations unrestricted.

=====

OVERALL RECOMMENDATION

Deploy immediately Thursday afternoon (3:00 PM–7:00 PM) if crew mobilized. Prioritize Friday and Monday for primary static sessions given superior PDOP and clear weekday logistics. Hold Wednesday 4:00 PM–8:00 PM as the highest-confidence backup window if weather or satellite geometry forces rescheduling early in the week. Extend all static occupations +1 hour minimum to compensate for residual ionospheric delay below normal; dual-frequency receivers and post-processing smoothing essential for accuracy.

=====

END REPORT

COUNTY RECORDING OFFICES

RECORDER / REGISTER OF DEEDS

Santa Cruz County Assessor-Recorder
701 Ocean St, Santa Cruz, CA 95060

ASSESSOR / TAX OFFICE

Santa Cruz County Assessor
701 Ocean St, Santa Cruz, CA 95060

This report compiles data from public and third-party sources (NGS, FEMA, USGS, NOAA, and county records) current as of the generation date shown above. GNSS planning values are predictive and conditions may change. This document is provided for planning purposes only and does not constitute a boundary survey, legal land description, or engineering certification. Verify all critical values against primary sources.

CORS STATIONS – WITHIN 30 MILES

QR / LINK	STATION ID	NAME / LOCATION	DISTANCE
 NGS ↗	P534	COOKEPEAK_CN2007	13.1 mi
 NGS ↗	P210	ELKHNSLGHNCN2005	19.6 mi
 NGS ↗	P217	LACROSSEDRCN2005	23.7 mi
 NGS ↗	P277	PIGEONPT__CN2007	24.3 mi
 NGS ↗	P231	HOPKINSSTNCN2006	24.4 mi
 NGS ↗	PPT6	PIGEON POINT 6	25.0 mi

NGS BENCHMARKS – WITHIN 10 MILES

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
 Datasheet ↗	GU1954	"V"	17.526m / 57.5ft NAVD88	Order 1	0.2 mi
 Datasheet ↗	GU1955	"U"	18.570m / 60.9ft NAVD88	Order	0.2 mi
 Datasheet ↗	GU4240	"M"	17.293m / 56.7ft NAVD88	Order 1	0.4 mi
 Datasheet ↗	GU1950	"U"	3.505m / 11.5ft NAVD88	Order	0.7 mi
 Datasheet ↗	GU1951	"V"	4.690m / 15.4ft NAVD88	Order	0.7 mi
 Datasheet ↗	GU1959	"W"	21.759m / 71.4ft NAVD88	Order 1	0.8 mi
 Datasheet ↗	GU1949	"RV 1 SPC0"	3.790m / 12.4ft NAVD88	Order	0.8 mi
 Datasheet ↗	GU1945	"14"	5.073m / 16.6ft NAVD88	Order 1	0.8 mi

QR / LINK	PID	DESIGNATION	ELEVATION	ORDER	DISTANCE
Datasheet					

FEMA FLOOD ZONE

X – AREA OF MINIMAL FLOOD HAZARD

Panel: 06087C0333F

Effective: Sep 29, 2017

[View FEMA Map ↗](#)



DATA COLLECTOR IMPORT – NGS BENCHMARKS

NGS BENCHMARK CONTROL POINTS – Copy / Paste to Data Collector

Datum: NAD83 | Vertical: NAVD88 | Source: NGS NDE API

PID	NAME	LAT	LON	ELEV_FT	ORDER
GU1954	V	36.961142	-122.041683	57.5	1
GU1955	U	36.961389	-122.041389	60.9	
GU4240	M	36.963611	-122.035556	56.7	1
GU1950	U	36.963889	-122.027778	11.5	
GU1951	V	36.963889	-122.027500	15.4	
GU1959	W	36.959050	-122.052233	71.4	1
GU1949	RV 1 SPC0	36.962500	-122.025000	12.4	
GU1945	14	36.964722	-122.026111	16.6	1

[Download .txt File](#)

USGS STREAM GAUGES – within 25 miles

STATION	DIST	LEVEL	FLOW	FLOOD STATUS
SAN LORENZO R A SANTA CRUZ CA #11161000	2.2 mi	1.85 ft	19.3 cfs	● Normal
SOQUEL C A SOQUEL CA #11160000	5.1 mi	2.36 ft	5 cfs	● Normal
SAN LORENZO R A BIG TREES CA #11160500	6.2 mi	3.29 ft	22.9 cfs	● Normal
APTOS C NR APTOS CA #11159690	7.9 mi	—	—	● Normal
CORRALITOS C A FREEDOM CA #11159200	14.9 mi	3.01 ft	0.14 cfs	● Normal
PAJARO R A WATSONVILLE CA #11159500	16.2 mi	12.06 ft	10.4 cfs	— Unknown
SALSIPUEDES C NR WATSONVILLE CA #11159490	17.1 mi	—	—	— Unknown
SARATOGA C A SARATOGA CA #11169500	20.4 mi	2.06 ft	1.1 cfs	— Unknown

CALIFORNIA RESERVOIRS (DWR/CDEC) – within 150 miles

RESERVOIR	DIST	ELEVATION	FULL POOL	STORAGE	% CAP
Lake Del Valle	47.8 mi	696.62 ft (-9 ft)	706 ft	36K af	67%
San Luis Reservoir	53.2 mi	467 ft (-110 ft)	577 ft	—	—
Don Pedro Reservoir	103.3 mi	811.33 ft (-19 ft)	830 ft	1,799K af	89%
Lake McClure	106.8 mi	827.26 ft (-40 ft)	867 ft	770K af	75%
New Melones Lake	107.5 mi	1,019.43 ft (-69 ft)	1,088 ft	1,657K af	69%
Millerton Lake	129.9 mi	636.59 ft (+59 ft)	578 ft	53K af	10%
Folsom Lake	129.9 mi	446.18 ft (-20 ft)	466 ft	767K af	79%

ARMY CORPS RESERVOIRS (USACE) – within 150 miles

RESERVOIR	DIST	POOL ELEV	FULL POOL	STORAGE	% CAP
Del Valle	50 mi	696.6 ft	695.01 ft	—	100%
Farmington Dam-Pool	101 mi	124.2 ft	117.09 ft	—	106%
Camanche	113 mi	221.86 ft*	221.86 ft	—	—
Burns-Pool	125 mi	265.9 ft	267.05 ft	—	100%
Bear-Pool	127 mi	344.4 ft	344.95 ft	—	100%
Eastman Lake-Pool	143 mi	484.8 ft	508.76 ft	—	95%
Hensley lake-Pool	149 mi	471.6 ft	495.29 ft	—	95%

USGS GROUNDWATER WELLS – within 15 miles

STATION	DIST	WELL DEPTH	WATER LEVEL	DATE	AQUIFER
012S001E23R001M	13.7 mi	300 ft	0.00 ft BGS	1975-04-24	N100CACSTL
012S001E25F001M	14.2 mi	604 ft	22.40 ft BGS	1983-09-08	N100CACSTL
012S001E24Q001M	14.2 mi	557 ft	24.01 ft BGS	1983-09-08	N100CACSTL
012S001E25B003M	14.3 mi	580 ft	18.72 ft BGS	1983-08-09	N100CACSTL
012S001E25G001M	14.5 mi	168 ft	17.95 ft BGS	1983-09-08	N100CACSTL
012S001E25B001M	14.5 mi	—	0.35 ft BGS	1983-03-11	N100CACSTL
012S002E30M002M	15.1 mi	—	—	—	N100CACSTL
012S002E30N001M	15.2 mi	—	—	—	N100CACSTL

AI
AQUIFER INTELLIGENCE
—
CLAUDE HAIKU

The Santa Cruz Mudstone (N100CACSTL) is a fine-grained marine sedimentary formation deposited during the Oligocene to Pliocene epochs along the California Coast Ranges. Its low permeability and limited interconnected porosity make it a poor aquifer; water occurs mainly in localized fractures and weathered zones near the surface or in thin intercalated sand layers. At this site, expect limited sustainable yield and highly variable water availability depending on proximity to structural breaks and permeable interbeds.

FORMATION

Name	Santa Cruz Mudstone
Age / Rock	Oligocene to Pliocene marine mudstone and siltstone with minor sand interbeds.
Extent	800–1200 sq mi (est. USGS/DWR)

DEPTH & RECHARGE

Depth Range	168–604 ft (17–60 floors)
Typical Depth	200–400 ft (20–40 floors)
Recharge Source	Precipitation on outcrop areas and lateral flow from higher-elevation sedimentary and metamorphic uplands
Transit Time	30–100 years to 300 ft depth (est. USGS/DWR)

WATER QUALITY

pH	est. 6.8–7.6 (est. USGS/DWR)
Hardness	est. 250–450 mg/L CaCO3 (est. USGS/DWR)
TDS	est. 350–700 mg/L (est. USGS/DWR)
Notes	No sample data — typical for formation; marine ancestry suggests potential for localized brackish water at depth

TREND

Trend	Sharp decline from 0.00 ft BGS (April 1975) to 17.95–24.01 ft BGS (September 1983); indicates 18–24 ft cumulative drawdown over 8 years, consistent with pumping stress in a low-yield mudstone aquifer
Source	Historical water-level record (1975–1983); est. USGS/DWR regional decline rates
Seasonal Swing	6–24 ft (evident in 1983 data: March 0.35 ft vs. August–September 17.95–24.01 ft)

Source: USGS National Water Information System · Search all wells in this area → USGS NWIS Mapper ↗ · Water level data pending USGS API migration for CA state well numbers

LAND SUBSIDENCE & AQUIFER COMPACTION

✓
LOW SUBSIDENCE RISK
—
No known critically overdrafted basin

Source: CA DWR SGMA Bulletin 118 2024.1 · California only - full US pending server migration

AI
SUBSIDENCE ASSESSMENT
—
CLAUDE HAIKU

The area near 36.9588, -122.0384 lies outside California's critically overdrafted groundwater basins and is underlain by the N100CACSTL aquifer system, which has not experienced significant sustained groundwater overdraft historically. The somewhat poorly drained soil conditions and absence of documented intensive agricultural or municipal pumping in this region mean that aquifer-system compaction from groundwater withdrawal has not occurred at rates sufficient to produce measurable land subsidence. For surveyors working in this area today, historical elevation control and benchmarks are expected to remain stable and reliable for conventional surveying work.

SUBSIDENCE ASSESSMENT

Susceptibility	Low
Primary Mechanism	None - consolidated terrain
Annual Rate	<1 mm/yr (est. USGS)
Historic Loss	None documented
Aquifer Compaction	None - aquifer not over-pumped
SGMA Status	Not applicable - not in SGMA basin

BENCHMARK IMPLICATIONS

Monument Risk	Low - monuments stable
Elevation Validity	Good - no active subsidence
GNSS Recommendation	Standard benchmark verification adequate

SURVEY ACTION

Required Action	No special action required
Dataset	CA DWR SGMA Bulletin 118 2024.1
Coverage	California critically overdrafted basins only - full US coverage pending

STRUCTURAL & ENGINEERING PARAMETERS

SEISMIC (ASCE 7-22)

Site Class	D (default)
Risk Category	II

WIND / SNOW / ICE / FROST (ASCE 7-22)

Basic Wind Speed	92 mph
Ground Snow Load	6.4 psf
Winter Wind (W2)	0.25
Ice Thickness	None
Frost Depth	6 in minimum

Brown's Formula · C=1 (Clay, Group D) · AFI=1°F-days · SF=1.25
Verify with local building official before construction

Source: ASCE 7-22 Hazard Tool · gis.asce.org

CLIMATE ZONE (IECC 2021): Zone 3C · Marine · Marine - mild year-round, coastal influence (SF, LA coast) County: Santa Cruz

○ AI CLIMATE SUMMARY — RESIDENTIAL DESIGN

This marine climate zone has mild winters and cool summers with persistent coastal winds, so your building envelope needs robust air sealing and impact-resistant glazing rated for the 92 mph wind loads, particularly on west and north-facing elevations. Concentrate windows on the east and south sides to capture winter sun for passive heating while minimizing heat gain from afternoon west-facing exposure, and avoid large glass areas on the windward side where wind-driven rain pressure is highest. Your key passive move here is strategic cross-ventilation through operable windows on opposite facades to leverage the reliable coastal breezes for summer cooling without air conditioning, which will offset the high cost of seismic-resistant framing in this zone.

○ AI STRUCTURAL ASSESSMENT — CLAUDE HAIKU

This coastal marine site presents moderate wind exposure and low seismic hazard with somewhat poorly drained soils that may warrant careful foundation design and drainage planning. The Watsonville loam with poor drainage characteristics and shallow frost depth suggests conventional spread footings may be feasible, though subsurface investigation is anticipated to confirm bearing capacity and seasonal groundwater behavior. Wind loads at 92 mph are the primary structural driver for this location.

FOUNDATION

Type	Conventional spread footing anticipated, pending geotechnical confirmation of bearing capacity and drainage conditions
Basis	Watsonville loam (Group D, somewhat poorly drained) and gentle slopes suggest spread foundations may be suitable; however, seasonal groundwater and drainage characteristics may warrant deeper investigation or drainage mitigation. Consult a licensed geotechnical engineer to confirm.

STRUCTURAL IMPLICATIONS

Seismic	Seismic Design Category could not be determined from available data (Ss and S1 null). Ordinary detailing is anticipated for planning purposes, but engineer confirmation of site-specific seismic parameters is recommended before final design.
Wind Exposure	Exposure C (open terrain with scattered obstructions) suggested per ASCE 7-22 Risk Category II — verify site topography and surrounding features with engineer
Wind	Standard wind load design anticipated at 92 mph 3-second gust; no hurricane or extreme coastal wind exposure suggested for planning purposes
Snow	Light snow load (6.4 psf ground snow) and marine winter wind (W2=0.25) anticipated — standard roof design likely adequate; consult engineer for drift or exposure conditions

INVESTIGATIONS & INSPECTIONS

Geotech Required	Yes — geotechnical investigation is recommended to confirm Site Class D designation, bearing capacity, seasonal groundwater depth, and drainage behavior of Watsonville loam
Liquefaction Screen	Screening recommended if shallow groundwater is confirmed; Site Class D + somewhat poorly drained soils may warrant evaluation. Consult engineer on subsurface conditions.
Special Inspections	Standard inspections anticipated for Risk Category II construction; may increase if seismic parameters or foundation complexity warrant — confirm with engineer
Code Edition	ASCE 7-22 / IBC 2021 (planning reference only)
Site Class Note	Site Class D assumed for planning. Geotechnical investigation anticipated to confirm classification, bearing capacity, and groundwater conditions before final foundation design.

SOILS — USDA SOIL SURVEY

MAP UNIT:	Watsonville loam, thick surface, 0 to 2 percent slopes	SERIES:	Watsonville
DRAINAGE CLASS:	Somewhat poorly drained	HYDRO GROUP:	D
SOIL ORDER:	Mollisols	SUBORDER:	Albolls

○ AI SOIL INTELLIGENCE — CLAUDE HAIKU

Watsonville loam is a somewhat poorly drained Mollisol (Alboll) formed in alluvial material on nearly level valley floors and low terraces in the Pajaro Valley region. The soil exhibits a thick, dark surface layer typical of mollisols, with an albic (bleached) subsurface horizon indicating periodic water saturation and iron-manganese mobilization. Field personnel should expect a loamy upper profile transitioning to clay-rich subsoil with seasonal high water tables; moisture retention and slow internal drainage will influence foundation design, fill placement, and subsurface utility installation.

Alluvial parent material on low-relief valley bottomland; native vegetation adapted to imperfect drainage conditions.

DRAINAGE

Class	Somewhat poorly drained
Seasonal Saturation	Water table 18–36 inches during winter and spring months (December–April)
Flooding Risk	Low to minimal on 0–2% slopes; check FEMA flood maps and local drainage history
Ponding Risk	Minor — localized surface ponding possible in depressions after heavy rainfall

ENGINEERING

Bearing Capacity	2,000–3,500 psf (shallow foundations); reduce by 30–50% during wet season
Rating	Fair
Shrink-Swell	Moderate — clay substratum (montmorillonite); differential settlement likely with seasonal moisture fluctuation
Frost Action	Low
Cut Suitability	Fair (material is predominantly silt and clay; limited drainage for cut slopes)
Fill Suitability	Fair (clay loam to clay substratum — moisture conditioning and compaction control essential to avoid rutting and poor consolidation)

SEPTIC

Suitability	Conditional
Notes	Perc test required; clay substratum and seasonal saturation may limit conventional system performance. Alternative systems (mound, sand filter, engineered treatment) likely necessary. Consult county health department.

HAZARDS

Excavation	Moderate — clay-rich subsoil will be sticky and difficult to work when wet; dewatering may be necessary
Expansive Clay	Moderate — clay substratum shows evidence of iron-manganese mottling consistent with shrink-swell potential; monitor structural distress
Erosion	Low on 0–2% slopes; potential for surface erosion and gully formation in disturbed or graded areas with concentrated runoff
Other	High seasonal water table — confirm depth and timing with local well logs and drainage studies before design. Verify historical flooding and subsidence risk for the valley setting.

Data sources: USGS National Water Information System · [waterdata.usgs.gov](#) · California DWR / CDEC · [cdec.water.ca.gov](#) · USACE CWMS · [water.usace.army.mil](#) · USGS NWIS Groundwater · [waterservices.usgs.gov](#) · USDA Web Soil Survey · [sdmdataaccess.nrcs.usda.gov](#) · CA DWR SGMA Bulletin 118 · [gis.water.ca.gov](#) · ASCE 7-22 Hazard Tool · [gis.asce.org](#) · USGS Design Maps · [earthquake.usgs.gov](#) · IECC 2021 Climate Zones · DOE Building America · Real-time data subject to revision

 OIL & GAS WELLS California — within 10mi radius

13 **0** **13** By status:
total active plugged **13** Plugged

By type: 12 DH 1 CH

Lease / Well	Status	Type	Operator	Depth	Dist
Swanton Improvement Co.	Plugged	DH	Swanton Improvement Co.	—	0.9 mi
Danish Oil & Devl Co.	Plugged	DH	Danish Oil & Devl Co.	—	3.2 mi
Monterey Bay Oil Co. Ltd	Plugged	DH	Monterey Bay Oil Co. Ltd	—	3.3 mi
Greater Stanta Cruz Oil Corp.	Plugged	DH	Greater Stanta Cruz Oil Corp.	—	3.5 mi
Skookum Oil & Invest Corp.	Plugged	DH	Skookum Oil & Invest Corp.	—	4.3 mi
Exxon Mobil Corporation	Plugged	DH	Exxon Mobil Corporation	—	4.9 mi
Santos, Oneto & Fabry Exp	Plugged	DH	Santos, Oneto & Fabry Exp	—	5.6 mi
Santos, Oneto & Fabry Exp	Plugged	DH	Santos, Oneto & Fabry Exp	—	5.8 mi
Joseph Faria, Jr.	Plugged	DH	Joseph Faria, Jr.	—	6 mi
Donald & Alice Santos	Plugged	DH	Donald & Alice Santos	—	6.6 mi
Oneto Bros	Plugged	DH	Oneto Bros	—	6.9 mi
Shell Western E&P Inc.	Plugged	CH	Shell Western E&P Inc.	—	8.2 mi
B. J. Marks Family	Plugged	DH	B. J. Marks Family	—	11.5 mi

AI Assessment — Well Proximity

The nearest well is the Swanton Improvement Co. plugged dry hole at 0.9 miles, which is sealed and beyond the distance that requires action. With no active producers within 1.5 miles, no injection wells within 1 mile, and no idle wells within 0.5 miles, well proximity is not a planning concern for this site. You should still verify surface easements and any pipeline rights-of-way that may cross the property, as these can constrain future development regardless of well distance.

Source: California public well records. Active = currently producing or permitted. Plugged = abandoned/decommissioned. Data updated nightly.

200
total in search area

6
within 5 mi

90
within 25 mi

Site Name	Status	Commodities	Distance
Lone Star Industries Inc.	Past Producer	Limestone, Clay	1.7 mi
Unnamed Location	Occurrence	LST_D	2 mi
Twin Lakes Beach	Past Producer	Gold	2.2 mi
Twin Lakes Beach	Producer	Gold	2.2 mi
Gibson Sand Fit	Past Producer	Sand/Gravel	3 mi
Unnamed Location	Occurrence	Stone/Crushed	4.5 mi
Santa Cruz Sand Pit and Mill	Producer	Sand/Gravel	6.2 mi
Decomposed Granite Pit	Occurrence	GRT	6.3 mi
Aptos Placer	Producer	Iron, TI_M	6.9 mi
Olympia Pit	Producer	Sand/Gravel	7.6 mi
Geyer Pit	Producer	Sand/Gravel	7.6 mi
Unnamed Location	Occurrence	Stone/Crushed	7.8 mi
Ixl Quarry	Past Producer	LST_D	7.9 mi
San Vicente Canyon	Past Producer	Stone/Crushed, Clay	9.8 mi
San Vincente Quarry	Producer	Stone/Crushed	10 mi
Triumph Steel Co	Occurrence	Titanium	14.5 mi
Alma	Prospect	STN	14.9 mi
Unnamed Location	Occurrence	Gold	16 mi
Bowie	Occurrence	Mercury	16.1 mi
Waddell Creek	Occurrence	Gold	16.7 mi
Steinheimer Ranch Prospect	Prospect	Chromium	16.8 mi
Monterey Bay Salt Works	Producer	NA_S	17.3 mi
Brainard	Occurrence	Mercury	17.3 mi
Kilday Ranch Deposits	Producer	Chromium	17.7 mi
Moss Landing Magnesia Plant	Plant	Chromium, MG	17.7 mi

Showing 25 of 200 deposits. Full dataset via USGS MRDS.

AI Assessment — Mining Context

There is 1 active producer within 5 miles: Twin Lakes Beach (2.2 mi) for gold mining. Expect heavy truck traffic on access roads during operating hours and noise during mining operations. No inactive deposits warrant Phase 1 ESA consideration as none are located within 1 mile of the site.

Source: USGS Mineral Resources Data System (MRDS). Status: Producer = active/recent; Past Producer = historical; Prospect = explored not mined; Occurrence = mineralization noted.

ENDANGERED & THREATENED SPECIES - USFWS Critical Habitat within 25mi

6
Endangered

7
Threatened

13
Total w/ Critical Habitat

6 Endangered species with designated critical habitat. ESA Section 7 consultation may be required for federal nexus projects.

COMMON NAME	SCIENTIFIC NAME	STATUS
Robust spineflower	<i>Chorizanthe robusta</i> var. <i>robusta</i>	Endangered

COMMON NAME	SCIENTIFIC NAME	STATUS
Scotts Valley polygonum	<i>Polygonum hickmanii</i>	Endangered
Scotts Valley spineflower	<i>Chorizanthe robusta</i> var. <i>hartwegii</i>	Endangered
tidewater goby	<i>Eucyclogobius newberryi</i>	Endangered
Yadon's piperia	<i>Piperia yadonii</i>	Endangered
Zayante band-winged grasshopper	<i>Trimerotropis infantilis</i>	Endangered
Bay checkerspot butterfly	<i>Euphydryas editha bayensis</i>	Threatened
California red-legged frog	<i>Rana draytonii</i>	Threatened
California tiger Salamander	<i>Ambystoma californiense</i>	Threatened
Marbled Murrelet	<i>Brachyramphus marmoratus</i>	Threatened
Monterey spineflower	<i>Chorizanthe pungens</i> var. <i>pungens</i>	Threatened
Santa Cruz tarplant	<i>Holocarpha macradenia</i>	Threatened
western snowy plover	<i>Charadrius nivosus nivosus</i>	Threatened

AI SPECIES & ESA ASSESSMENT - CLAUDE HAIKU

ESA Implications Review – 36.9588, -122.0384 **Section 7 Consultation Trigger:** If this project involves federal permitting, funding, or licensing, Section 7 consultation with USFWS is required given the 13 species with critical habitat within 25 miles. **Most Planning-Relevant Species:** The terrestrial-focused endangered species (Robust spineflower, Scotts Valley polygonum/spineflower, Zayante band-winged grasshopper) and threatened species (Santa Cruz tarplant, Bay checkerspot butterfly) present the highest regulatory risk if site conditions support their habitat requirements; aquatic species (tidewater goby, California red-legged frog) merit screening if wetlands or streams exist on-site. **Recommended Next Steps:** Conduct a Phase 1 biological assessment to determine site-specific habitat presence, species occurrence likelihood, and critical habitat overlap; this will establish whether project design modifications, habitat surveys, or formal incidental take authorization are necessary. For planning purposes only. Consult a licensed environmental biologist for formal ESA compliance.

Source: USFWS Critical Habitat Final Designated Areas - FWS Open Data - Critical habitat does not equal species range

NATIVE PLANTS - iNaturalist Research-Grade within 10mi

52	51800	2026-07-15
Unique Species	Total Observations	Most Recent
COMMON NAME	SCIENTIFIC NAME	LAST OBS
American lyme grass	<i>Leymus mollis</i>	2026-07-13
American trailplant	<i>Adenocaulon bicolor</i>	2026-07-09
Arroyo Willow	<i>Salix lasiolepis</i>	2026-07-13
beach suncup	<i>Camissoniopsis cheiranthifolia</i>	2026-07-13
beaked hazelnut	<i>Corylus cornuta</i>	2026-07-09
bigleaf maple	<i>Acer macrophyllum</i>	2026-07-09
bishop pine	<i>Pinus muricata</i>	2026-07-14
California bay	<i>Umbellularia californica</i>	2026-07-12
California fetid adderstongue	<i>Scoliopus bigelovii</i>	2026-07-09
California Harebell	<i>Smithiastrum prenanthoides</i>	2026-07-08
California Hedge Nettle	<i>Stachys bullata</i>	2026-07-10
California incense-cedar	<i>Calocedrus decurrens</i>	2026-07-15
California mugwort	<i>Artemisia douglasiana</i>	2026-07-08
California poppy	<i>Eschscholzia californica</i>	2026-07-13
California Spikenard	<i>Aralia californica</i>	2026-07-08
California torreyia	<i>Torreya californica</i>	2026-07-08
coast live oak	<i>Quercus agrifolia</i>	2026-07-13
coast redwood	<i>Sequoia sempervirens</i>	2026-07-13
coastal bush lupine	<i>Lupinus arboreus</i>	2026-07-13
coffeeberry	<i>Frangula californica</i>	2026-07-11
common selfheal	<i>Prunella vulgaris</i>	2026-07-15
creeping snowberry	<i>Symphoricarpos mollis</i>	2026-07-13
Evergreen Huckleberry	<i>Vaccinium ovatum</i>	2026-07-09
fragrant bedstraw	<i>Galium triflorum</i>	2026-07-13
Green pin-cushion alga	<i>Cladophora columbiana</i>	2026-07-12
lady fern	<i>Athyrium filix-femina</i>	2026-07-09
Monterey Monkeyflower	<i>Diplacus linearis</i>	2026-07-09
Narrowleaf Willow	<i>Salix exigua</i>	2026-07-13
orange bush monkeyflower	<i>Diplacus aurantiacus</i>	2026-07-12
Pacific False Bindweed	<i>Calystegia purpurata</i>	2026-07-10

COMMON NAME	SCIENTIFIC NAME	LAST OBS
Pacific hound's tongue	<i>Adelinia grandis</i>	2026-07-13
Pacific madrone	<i>Arbutus menziesii</i>	2026-07-15
Pacific poison oak	<i>Toxicodendron diversilobum</i>	2026-07-13
Pacific trillium	<i>Trillium ovatum</i>	2026-07-09
Pink Honeysuckle	<i>Lonicera hispidula</i>	2026-07-08
Pink Sand Verbena	<i>Abronia umbellata</i>	2026-07-12
Redwood Sorrel	<i>Oxalis smalliana</i>	2026-07-12
Redwood Violet	<i>Viola sempervirens</i>	2026-07-09
Scouring-pad alga	<i>Endocladia muricata</i>	2026-07-12
Seaside Buckwheat	<i>Eriogonum latifolium</i>	2026-07-14
seep monkeyflower	<i>Erythranthe guttata</i>	2026-07-09
Stream Orchid	<i>Epipactis gigantea</i>	2026-07-08
tall flatsedge	<i>Cyperus eragrostis</i>	2026-07-09
Tanoak	<i>Notholithocarpus densiflorus</i>	2026-07-09
thimbleberry	<i>Rubus parviflorus</i>	2026-07-09
tomcat clover	<i>Trifolium willdenovii</i>	2026-07-13
western blue-eyed grass	<i>Sisyrinchium bellum</i>	2026-07-09
western maidenhair fern	<i>Adiantum aleuticum</i>	2026-07-09
western sword fern	<i>Polystichum munitum</i>	2026-07-12
Western Wild Ginger	<i>Asarum caudatum</i>	2026-07-09
wing-leaf monkeyflower	<i>Erythranthe ptilota</i>	2026-07-09
yerba buena	<i>Clinopodium douglasii</i>	2026-07-15

AI PLANT COMMUNITY SUMMARY - CLAUDE HAIKU

Ecological Assessment: 36.9588, -122.0384 This site supports a **coastal redwood forest and oak woodland mosaic**, typical of the Santa Cruz Mountains transition zone where coast redwoods intermix with coast live oak and Douglas-fir associates. The presence of moisture-loving species (arroyo willow, California spikenard, redwood) alongside drought-tolerant chaparral elements (bishop pine, coastal bush lupine, California poppy) indicates a site with **variable microtopography and seasonal water availability**—likely featuring seeps or seasonal drainages interspersed with drier ridges. Soil moisture will be high in valley bottoms and north-facing slopes (potentially problematic for conventional grading without subsurface drainage), while ridge and south-facing areas will drain more rapidly; plan grading around existing drainage patterns rather than concentrating water flow. The dense native plant diversity suggests intact soil structure and mycorrhizal networks that will be disrupted by compaction, so minimize disturbance footprints and avoid heavy equipment on slopes steeper than 15%. Consult a licensed botanist for site-specific vegetation assessment.

Source: iNaturalist research-grade observations - Native species filter applied

FIELD HAZARDS - Venomous & Toxic Species within 25mi

242	199	150	3092
Venomous Snakes	Venomous Spiders	Stinging Insects	Toxic Plants
VENOMOUS SNAKES			
SPECIES		OBSERVATIONS	
Crotalus oreganus oreganus		242 obs	
Crotalus oreganus		242 obs	
VENOMOUS SPIDERS			
SPECIES		OBSERVATIONS	
Latrodectus hesperus		199 obs	
Latrodectus geometricus		199 obs	
STINGING INSECTS			
SPECIES		OBSERVATIONS	
Dasymutilla sackenii		150 obs	
Dasymutilla aureola		150 obs	
Dasymutilla californica		150 obs	
Dasymutilla flammifera		150 obs	
TOXIC PLANTS			
SPECIES		OBSERVATIONS	
Toxicodendron diversilobum		1967 obs	
Datura stramonium		86 obs	
Brugmansia aurea		86 obs	

SPECIES	OBSERVATIONS
<i>Datura wrightii</i>	86 obs
<i>Datura discolor</i>	86 obs
<i>Conium maculatum</i>	889 obs
<i>Urtica gracilis</i>	150 obs
<i>Urtica dioica</i>	150 obs
<i>Urtica gracilis holosericea</i>	150 obs

AI FIELD SAFETY BRIEFING - CLAUDE HAIKU

Field Safety Briefing: Monterey Bay Area (36.9588, -122.0384) **Highest-Priority Hazards:** Your greatest risks are toxic plants—particularly poison oak (*Toxicodendron diversilobum*) with 3,092 observations—and venomous snakes (*Crotalus oreganus*, the Northern Pacific rattlesnake) which are common in this coastal scrubland. Black widows and other venomous spiders are secondary concerns but present. **Practical Avoidance:** Wear long pants, long sleeves, and closed-toe boots; learn to identify poison oak (three leaflets, oily residue) and avoid touching it or contaminated clothing. Stay alert for rattlesnakes in brush and rocky areas—they typically retreat if given space, so make noise while walking and watch your hand placements. Inspect work areas before kneeling or sitting. **Seasonal Considerations:** Rattlesnakes are most active April through October when temperatures rise; poison oak remains a hazard year-round but causes worse reactions in spring/summer. Stinging insects peak in warmer months. Plan work in cooler months when possible and brief crew on recognizing nest sites. Always verify current conditions locally before fieldwork.

Source: iNaturalist research-grade observations - Always verify locally before fieldwork

FAA AIRWAYS — ATS Routes crossing site area

19	12	3	4			
Total Airways	Jet Routes	Victor Airways	Other Routes			
AIRWAY	TYPE	LEVEL	MEA EAST	MEA WEST	TRACK	REV TRACK
J1	Jet Route	High	18,000 ft	—	303.8°	121.4°
J110	Jet Route	High	18,000 ft	—	137.9°	318.2°
J126	Jet Route	High	18,000 ft	—	308.1°	127.9°
J3	Jet Route	High	18,000 ft	—	342.8°	161.8°
J32	Jet Route	High	18,000 ft	—	19.2°	199.6°
J501	Jet Route	High	18,000 ft	—	301.5°	121.1°
J58	Jet Route	High	18,000 ft	—	65.3°	245.9°
J6	Jet Route	High	18,000 ft	—	127.8°	307.9°
J80	Jet Route	High	18,000 ft	—	65.3°	245.9°
J84	Jet Route	High	18,000 ft	—	50.8°	231.0°
J88	Jet Route	High	18,000 ft	—	308.1°	127.9°
J94	Jet Route	High	18,000 ft	—	65.3°	245.9°
T257	T-Route	Low	—	—	322.0°	141.7°
T259	T-Route	Low	—	—	41.6°	221.8°
T261	T-Route	Low	—	—	31.4°	211.7°
T263	T-Route	Low	—	—	308.8°	128.2°
V107	Victor Airway	Low	7,000 ft	—	295.4°	114.2°
V109	Victor Airway	Low	4,000 ft	—	240.1°	60.0°
V111	Victor Airway	Low	5,500 ft	—	11.0°	191.0°

Source: FAA ATS Route FeatureServer — Airways shown cross the site bounding area and may not pass directly overhead — MEA = Minimum En Route Altitude

RAILROAD PROXIMITY — FRA North American Rail Network

7						
Rail Lines						
OWNER / OPERATOR	TYPE	TRACKS	PASSENGER	STRACNET	SUBDIVISION	
SCBG	X	1	No	No	SAN FRANCISCO	
SPP	X	1	No	No	COASTAL SUBDIVISION	
SCBG	Mainline	1	No	No	COASTAL SUBDIVISION	
SCBG	Other	1	No	No	—	
SCBG	S	1	No	No	—	
SPP	Mainline	1	No	No	COASTAL SUBDIVISION	
Unknown	A	1	No	No	—	

Source: FRA North American Rail Network (NARN) — Updated July 2025 — STRACNET = Strategic Rail Corridor Network (DOD)

WILDLIFE & NATURAL HISTORY - iNaturalist within 25mi

200	88	64	13	10	1
Total Observations	Total Species	Birds	Mammals	Reptiles	Amphibians

BIRDS (64 species)

COMMON NAME	Scientific Name	OBS
Rock Pigeon	<i>Columba livia</i>	12
Feral Pigeon	<i>Columba livia domestica</i>	12
Western Gull	<i>Larus occidentalis</i>	8
Wild Turkey	<i>Meleagris gallopavo</i>	7
California Towhee	<i>Melospiza crissalis</i>	4
Wood Duck	<i>Aix sponsa</i>	4
Great Horned Owl	<i>Bubo virginianus</i>	3
Turkey Vulture	<i>Cathartes aura</i>	3
Heermann's Gull	<i>Larus heermanni</i>	3
Cooper's Hawk	<i>Astur cooperii</i>	3
California Gull	<i>Larus californicus</i>	3
Red-necked Phalarope	<i>Phalaropus lobatus</i>	3
Common Murre	<i>Uria aalge</i>	3
Pigeon Guillemot	<i>Cephus columba</i>	3
California Quail	<i>Callipepla californica</i>	2

+ 49 additional species

MAMMALS (13 species)

COMMON NAME	Scientific Name	OBS
Eastern Pacific Long-beaked Common Dolphin	<i>Delphinus delphis bairdii</i>	14
Humpback Whale	<i>Megaptera novaeangliae</i>	8
California Ground Squirrel	<i>Otospermophilus beecheyi</i>	6
Common Bottlenose Dolphin	<i>Tursiops truncatus</i>	3
California Sea Lion	<i>Zalophus californianus</i>	2
Mule Deer	<i>Odocoileus hemionus</i>	2
Columbian Black-tailed Deer	<i>Odocoileus hemionus columbianus</i>	2
Coyote	<i>Canis latrans</i>	2
Harbor Seal	<i>Phoca vitulina</i>	2
Southern Sea Otter	<i>Enhydra lutris nereis</i>	1
Eastern Gray Squirrel	<i>Sciurus carolinensis</i>	1
Pacific Harbor Seal	<i>Phoca vitulina richardii</i>	1
Long-tailed Weasel	<i>Neogale frenata</i>	1

REPTILES (10 species)

COMMON NAME	Scientific Name	OBS
Western Fence Lizard	<i>Sceloporus occidentalis</i>	6
Coast Garter Snake	<i>Thamnophis elegans terrestris</i>	4
Pacific Gopher Snake	<i>Pituophis catenifer catenifer</i>	2
Santa Cruz Aquatic Garter Snake	<i>Thamnophis atratus atratus</i>	1
California King Snake	<i>Lampropeltis californiae</i>	1
Western Yellow-bellied Racer	<i>Coluber constrictor mormon</i>	1
Northern Alligator Lizard	<i>Elgaria coerulea</i>	1
Forest Sharp-tailed Snake	<i>Contia longicauda</i>	1
Southern Alligator Lizard	<i>Elgaria multicarinata</i>	1
Coast Range Fence Lizard	<i>Sceloporus occidentalis bocourtii</i>	1

AMPHIBIANS (1 species)

COMMON NAME	Scientific Name	OBS
Arboreal Salamander	<i>Aneides lugubris</i>	1

AI WILDLIFE & ECOLOGICAL ASSESSMENT - CLAUDE HAIKU

Ecological Character Assessment The site at 36.9588, -122.0384 represents a coastal transition zone with mixed terrestrial and marine habitats, supporting a diverse community of 88 species including both generalist urban-adapted birds (Rock Pigeon, Feral Pigeon) and sensitive coastal species (Western Gull, Heermann's Gull, Southern Sea Otter). The presence of marine

mammals (dolphins, whales, harbor seals, sea lions) and coastal-dependent raptors (Great Horned Owl, Cooper's Hawk) indicates proximity to productive marine or estuarine waters, while the terrestrial reptile and mammal assemblage suggests adjacent coastal scrub, oak woodland, or grassland habitat with varied microclimates supporting both aquatic (Santa Cruz Aquatic Garter Snake) and terrestrial species. Species of regulatory concern include the Southern Sea Otter (threatened under the ESA), marine mammals protected under the MMPA, and state-listed sensitive reptiles (Forest Sharp-tailed Snake), indicating that this area likely encompasses critical habitat corridors and marine protected areas requiring sensitive management. The ecological community suggests moderate to high disturbance tolerance with presence of invasive species (Eastern Gray Squirrel, Feral Pigeon) alongside native specialists, indicating a landscape mosaic of preserved natural habitats and human-influenced areas; any development should account for sensitive coastal access points, marine mammal movement corridors, and rare reptile habitat connectivity. **For planning purposes only. Consult a qualified biologist for site-specific biological assessments.**

Source: iNaturalist research-grade observations — Data reflects community science observations and may not represent complete species inventory

✳ PASSIVE + ACTIVE SOLAR PLANNING — Lat 36.96°

1 • AVERAGE SUN HOURS

Period	Peak Hrs/Day	Notes
Annual average	6.7 hrs	Overall solar resource
Summer average	9.1 hrs	June–August
Winter average	4 hrs	December–February
Worst month (Dec)	3.3 hrs	Size battery storage to this

2 • IDEAL SOLAR ARRAY PITCH

Optimal tilt = 37° (latitude) → 9/12 pitch

Pitch	Angle	% of Optimal	Notes
4/12	18.4°	97.2%	good
5/12	22.6°	97.8%	good
6/12	26.6°	98.4%	← excellent
7/12	30.3°	99%	← excellent
8/12	33.7°	99.5%	← excellent
9/12	36.9°	100%	← annual optimal
10/12	39.8°	99.6%	← excellent
12/12	45°	98.8%	← excellent

Going from 5/12 to 6/12 at framing costs ~\$800–1,200 one time. On a 10kW system that's +734 kWh/yr (~\$110 value/yr). Over 25yr panel life: ~\$2751 total gain.

3 • SOLAR HEAT GAIN — BTU/HR PER SQ FT

Use these values as inputs for eQUEST, EnergyPlus, Manual J, or HERS rating

Surface	Winter	Summer	Design Note
South glass (SHGC 0.4)	54	31	Minimize south glass
South glass (SHGC 0.6)	81	46	Clear/uncoated glass
North glass (any)	10	13	Negligible — diffuse only
East / West glass	29	105	△ Minimize — afternoon load
Roof — dark (absorb 0.95)	241	301	Highest load surface
Roof — light (absorb 0.35)	89	111	Cool roof — significant reduction
Roof — metal (absorb 0.25)	63	79	Best roof option for heat
South wall — dark	100	69	
South wall — light	33	23	Light color = major savings
Hardscape — asphalt	181	301	△ Heat island — avoid near structure
Hardscape — concrete	124	206	Also reflects onto adjacent walls
Hardscape — light pave	67	111	Recommended within 50ft
Pool surface	23	38	Low absorb + evaporative cooling

Concrete reflection onto south wall: ~4 BTU/hr-ft² additional load per 100sf of adjacent hardscape.

Pool specular reflection: angle-dependent — east/west pools can cause morning/afternoon glare into glass.

4 • ROOF PITCH SOLAR LOAD — BTU/HR PER 100 SQ FT (SUMMER)

Pitch	Dark Roof	Light Roof	Reduction vs Flat
0/12	301 BTU	111 BTU	baseline (worst)
4/12	241 BTU	89 BTU	-20% dark / -20% light
6/12	196 BTU	72 BTU	-35% dark / -35% light
8/12	166 BTU	61 BTU	-45% dark / -45% light
12/12	135 BTU	50 BTU	-55% dark / -55% light

Scale: multiply BTU × (your roof sf ÷ 100). Pitch + light color = biggest single cooling move.

5 • THERMAL TIME LAG — WALL ASSEMBLY

Peak outdoor temp ~15:00. Heat arrives at interior = peak + lag hours.

Assembly	Lag	Peak Interior Load	Note
2x4 frame (R-15)	0.6 hrs	~15:36 PM	R-value only — no mass benefit
2x6 frame (R-23)	0.9 hrs	~15:54 PM	R-value only — no mass benefit
2x8 frame (R-30)	1.2 hrs	~16:12 PM	R-value only — no mass benefit
2x10 frame (R-37)	1.5 hrs	~16:30 PM	good
2x12 frame (R-44)	1.8 hrs	~16:48 PM	good
ICF 6" core	7 hrs	~22:00 PM	✓ shifts past sunset — natural vent window
8" CMU block	6 hrs	~21:00 PM	✓ shifts past sunset — natural vent window
12" concrete	8.5 hrs	~23:30 PM	✓ shifts past sunset — natural vent window

In 3C Marine: ICF or CMU shifts peak load past sunset. Open windows, flush heat naturally. AC load drops significantly.

6 • TREE SHADOW PLANNING — TREE DUE SOUTH, 100FT FROM STRUCTURE

Tree Ht	Shadow Reaches	Summer hrs/day	Equinox hrs/day	Winter hrs/day	WUI note
10ft	All year	1.1	1	1.1	OK
20ft	All year	2.1	1.9	2.3	OK

30ft	All year	3	2.8	3.6	Zone 2 check
40ft	All year	3.9	3.7	5	△ verify
50ft	All year	4.7	4.5	6.7	△ verify
60ft	All year	5.5	5.3	9.5	X Zone 1

Deciduous tree: winter solar penetration ~85% (leaves off) — best of both worlds in hot-dry climates.
Evergreen tree: winter penetration ~15% — good windbreak, poor solar strategy south of structure.
△ WUI zones: tree placement must be coordinated with defensible space requirements. Verify with local fire authority — WUI boundaries vary by jurisdiction.

7 · ATTIC VENTING — PER 100 SQ FT OF ROOF AREA

Standard	Unbalanced (1/150)	Balanced† (1/300)	Notes
Code minimum	8 sq in	4 sq in	Moisture protection only
Thermal comfort	24 sq in	12 sq in	Keeps attic within 20°F of outdoor
WUI mesh 1/16"	60 sq in	30 sq in	2.5× compensation — mesh clogs
WUI intumescent	24 sq in	12 sq in	Self-closing — no compensation needed

Pitch factor: 0/12×1.0 · 4/12×0.80 · 6/12×0.65 · 8/12×0.55 · 12/12×0.45
Color factor: Dark×1.0 · Medium×0.75 · Light×0.55 · Metal×0.45
Scale: multiply sq in × (your roof sf ÷ 100) × pitch factor × color factor
†**Balanced** = ≥40% NFA within top third of roof height (ridge) + ≥40% within bottom third (soffit)
WUI conflict: Soffit vents are ember collection points. In high WUI zones — eliminate soffit vents, use ridge-only (unbalanced/1/150). In low WUI — balanced system preferred.
△ Verify WUI zone with local fire authority — boundaries vary significantly by jurisdiction.

🕒 AI SOLAR & PASSIVE DESIGN SUMMARY — HOMEOWNER GUIDE

Your Home's Solar & Energy Plan You're in a sweet spot for solar—you get about 6.7 hours of good sunlight per day on average, which is solid, and that jumps to 9 hours in summer when you need it most. Now, here's where most people miss an opportunity: your roof color and angle matter *hugely*. A dark roof soaks up over 300 BTU per hour per square foot in summer (think of it like leaving your car in the sun with closed windows), but switching to a light color cuts that nearly in thirds—and your roof's angle of about 37 degrees is nearly perfect for both solar panels and natural light. The real troublemaker is your west-facing windows, which blast your home with over 100 BTU per hour in summer afternoon heat, so consider shading those aggressively, maybe with an awning or shade cloth. Here's the smart move: plant a deciduous tree on your south side, about 100 feet away or closer depending on your lot—it'll give you nearly 4 hours of shade in summer when you need cooling, but drop its leaves in winter to let that warming sun through. Thermal mass is basically your home's ability to soak up heat during the day and release it slowly at night, and your framing type directly affects how well that works. These numbers are planning estimates—your architect or energy modeler can plug them directly into their software for a full analysis.

Planning estimates only — clear-sky model, standard assumptions. Values formatted for eQUEST / EnergyPlus / Manual J input. Verify with licensed architect, energy modeler, or solar installer before specifying.

Technical Brief — for architects, engineers, and surveyors

SITE INTELLIGENCE BRIEFING **36.95882°N, 122.03836°W | Santa Cruz County, CA**

Structural Design Drivers & Climate Envelope

This site falls within IECC Climate Zone 3C (Marine) with an exceptionally low annual freeze index of 1°F-day, effectively eliminating freeze-thaw as a structural design driver. Wind and snow load data are not available for this location; without those parameters, seismic demand cannot yet be confirmed as the controlling load case. The design team must obtain site-specific wind speed data (3-second gust) and snow load mapping from USGS or local building department records before finalizing the lateral load path. Given the marine microclimate designation and coastal Santa Cruz County location, wind exposure warrants priority investigation.

Geotechnical Foundation Requirements

The site is underlain by Watsonville loam with thick surface layer, 0–2 percent slopes, classified in Hydrologic Group D (poorly drained soils) with somewhat poor drainage characteristics. Group D soils exhibit high runoff potential and low infiltration; this drainage class typically demands one or more of the following: subsurface dewatering, perimeter drainage, or elevated finished grade. The 6-inch frost depth (derived from Brown's Formula using the 1°F-day AFI) is shallow and imposes minimal footing depth requirements; however, the low permeability of Group D soils may necessitate underdrains or French drains to maintain stability and reduce hydrostatic pressure on below-grade elements. Soil texture data are incomplete; a Phase I geotechnical investigation must include boring logs, Atterberg limits, and permeability testing to confirm bearing capacity and settlement risk.

Seismic & Subsidence Assessment

No Quaternary faults are mapped within the search radius of this site, and subsidence basin mapping shows no data for the area. The San Andreas Fault system lies well outside the immediate vicinity. While seismic activity cannot be ruled out for Santa Cruz County, the absence of nearby mapped faults reduces fault rupture hazard. Subsidence risk appears low based on available USGS basin delineations. Wildfire hazard data were not provided; local Cal Fire hazard mapping should be reviewed given Santa Cruz County's fire history.

Mining & Site Context

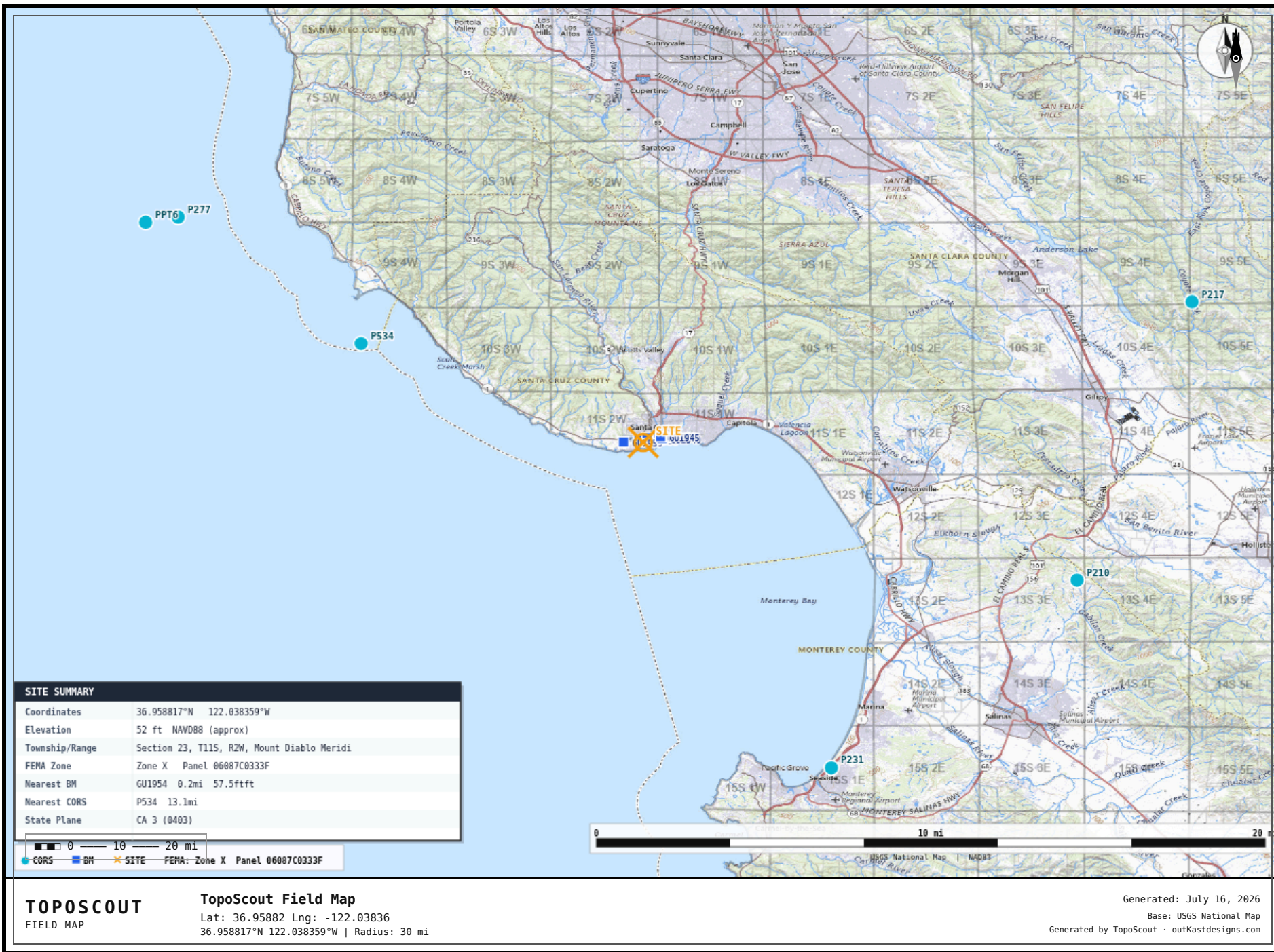
Approximately 200 MRDS (Mineral Resources Data System) deposits are recorded within a ~35-mile radius; however, only one active producer—Twin Lakes Beach (gold, 2.2 miles south)—operates within 5 miles of the site. This distance is sufficient to avoid direct mining subsidence or blast effects. Thirteen oil and gas wells are recorded within ~10 miles; the nearest is the Swanton Improvement Company well 0.9 miles away, which is plugged and inactive. No active oil or gas production occurs in the immediate area. These hazards present low risk to the proposed development.

Bottom Line

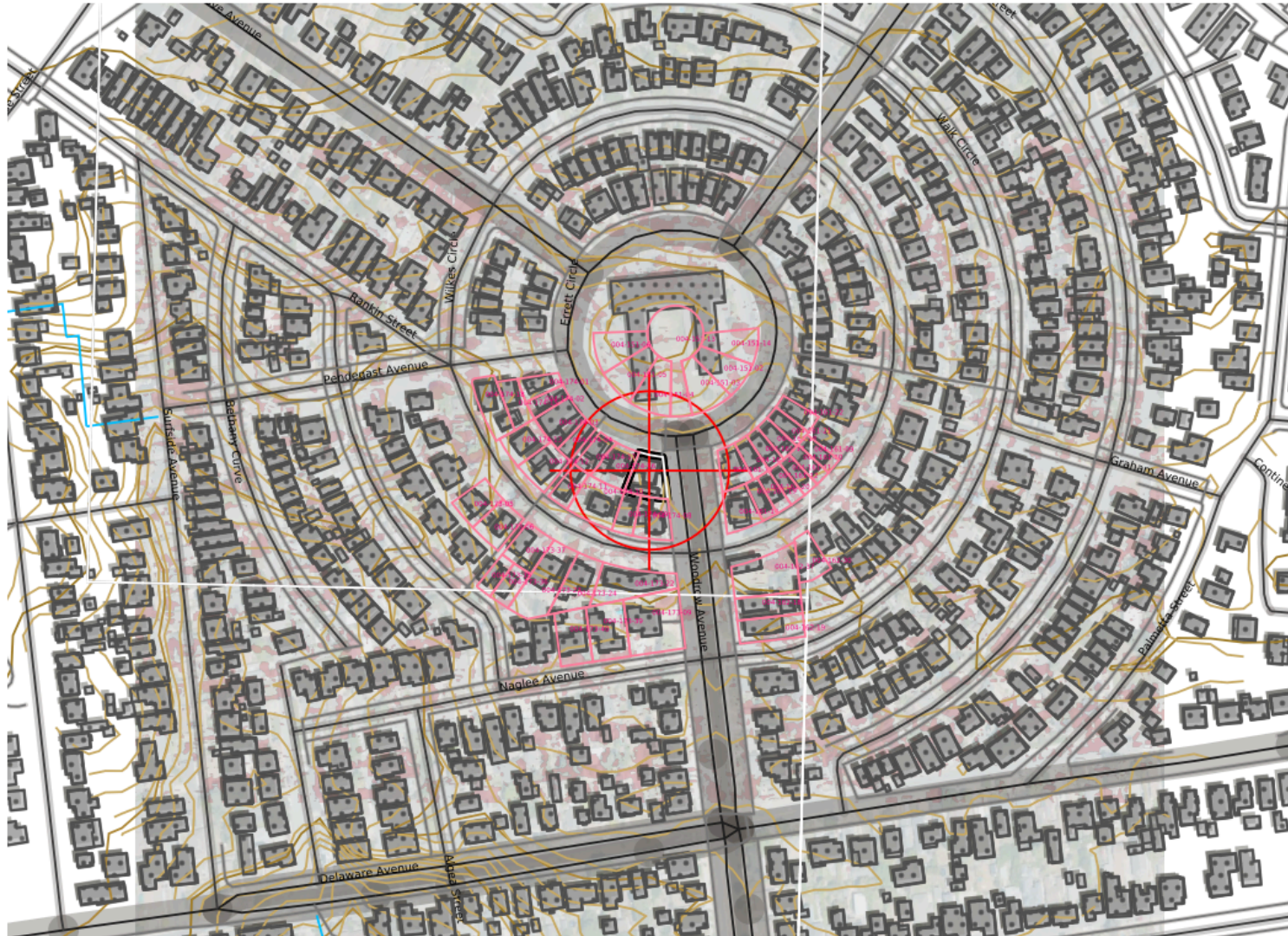
Obtain wind speed and snow load site-specific data from the building department and USGS, conduct Phase I geotechnical investigation with focus on Group D soil drainage and bearing capacity, and confirm no active subsidence or mining activity affects the site before finalizing foundation design.

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Planning aid only — not a substitute for licensed professional review, geotechnical investigation, or code compliance verification.

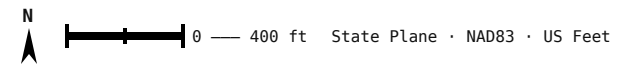


TOPOSCOUT SITE PLAN PREVIEW



INCLUDED LAYERS

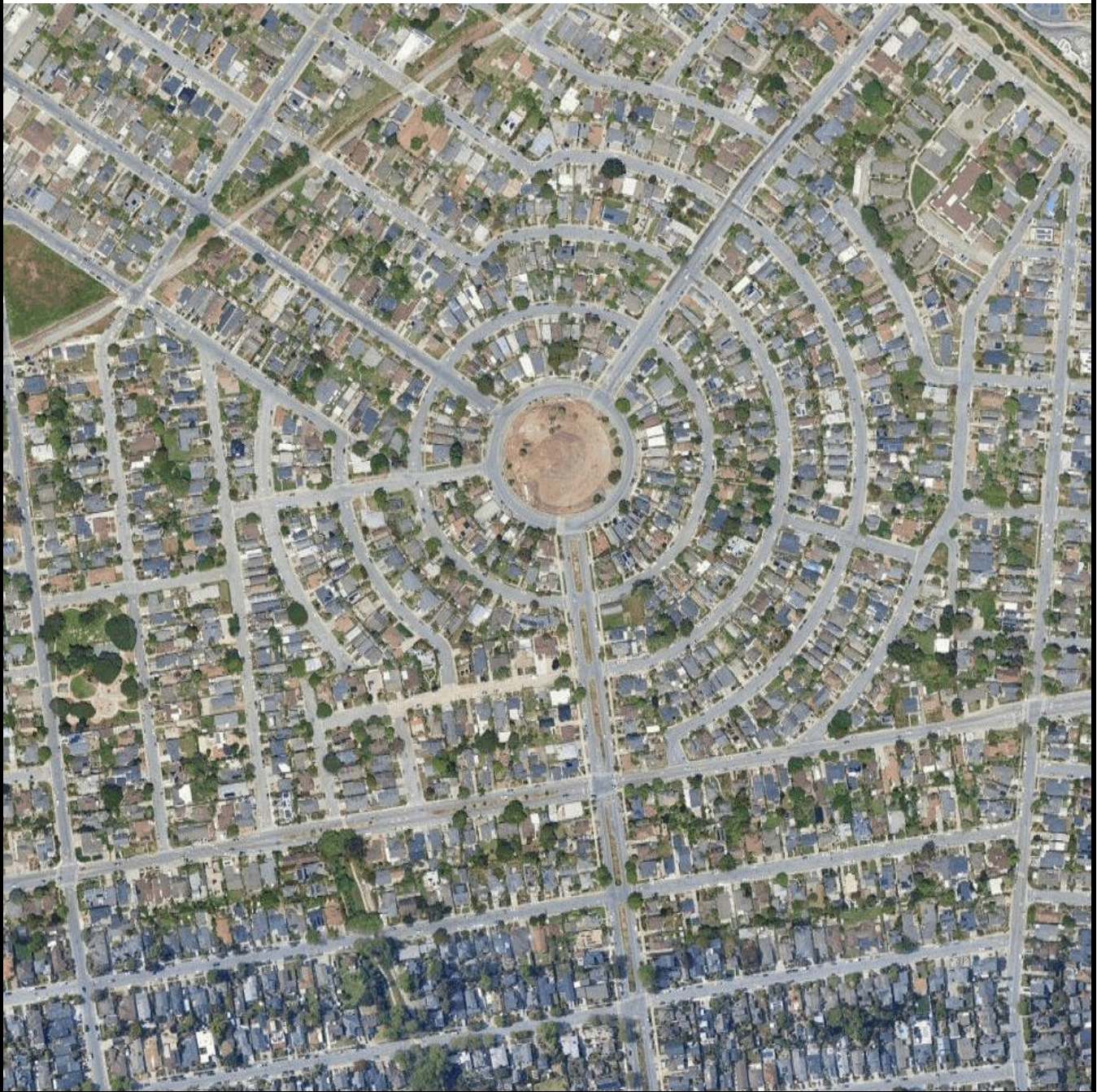
- Contours (Index/Major/Minor/1ft) ■ PLSS Section/Township/QQ ■ Roads Major/Local/Trail + ROW ■ Buildings ■ Hydro ■ Power Lines ■ Pipeline ■ Railroad ■ Benchmarks + Labels
- County Boundary ■ OTLS Grid ■ Site Info ■ Elevation Labels ■ 3D Polyline (all geometry)





SITE AERIAL – 600 ft × 600 ft

36.95882°N, 122.03836°W · Google Aerial



AREA CONTEXT — 3000 ft × 3000 ft

36.95882°N, 122.03836°W · Google Aerial

✂ FIELD RESOURCE LINKS

PRE-MISSION PLANNING

Trimble GNSS Planning Online –

<https://trimblegnssplanningonline.com/>

Satellite visibility, DOP charts, sky plots

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Drone airspace authorization, TFRs, restricted areas

Aloft (Kittyhawk) – <https://www.aloft.ai/>

Field-friendly drone airspace map and LAANC authorization

NOAA Space Weather — Live Kp –

<https://www.swpc.noaa.gov/products/planetary-k-index>

Real-time Kp index, geomagnetic storm alerts

POST-PROCESSING

NGS OPUS – <https://geodesy.noaa.gov/OPUS/index.jsp>

Online Positioning User Service – static GNSS processing

NGS OPUS-RS – <https://geodesy.noaa.gov/OPUSI/index.jsp>

Rapid static processing – sessions as short as 15 minutes

CORS Data Download – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Raw RINEX observation files from CORS stations

FEMA & FLOOD

FEMA Flood Map Service Center –

<https://msc.fema.gov/portal/home>

FIRM panels, flood zone lookup by address or coordinates

FEMA Elevation Certificate – [https://www.fema.gov/flood-](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

[insurance/work-with-nfip/elevation-certificate](https://www.fema.gov/flood-insurance/work-with-nfip/elevation-certificate)

Current EC form, instructions, LOMA/LOMR submission guidance

BASE STATION & NETWORK

NOAA CORS Network – [https://geodesy.noaa.gov/CORS/cors-](https://geodesy.noaa.gov/CORS/cors-data.shtml)

[data.shtml](https://geodesy.noaa.gov/CORS/cors-data.shtml)

Continuously operating reference stations, data download

NGS CORS Station Map –

<https://geodesy.noaa.gov/CORS/GoogleMap/CORSmap.shtml>

Interactive map of all CORS stations with status

IGS Station Monitor – <https://www.igs.org/network>

International GNSS Service – global base station health

REFERENCE & CALCULATIONS

NGS Datasheets – [https://www.ngs.noaa.gov/cgi-](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

[bin/datasheet.prl](https://www.ngs.noaa.gov/cgi-bin/datasheet.prl)

Benchmarks, control points, passive monuments

NGS Coordinate Conversion (NCAT) –

<https://geodesy.noaa.gov/NCAT/>

NADCON5, datum transforms, state plane conversions

NOAA Magnetic Declination –

<https://www.ngdc.noaa.gov/geomag/calculators/magcalc.shtml>

Declination calculator for any location and date

NGS Geoid Models – <https://geodesy.noaa.gov/GEOID/>

GEOID18 and current model downloads, GEOID calculator

DRONE OPERATIONS

FAA Part 107 — Commercial Rules –

https://www.faa.gov/uas/commercial_operators

Full Part 107 regulatory summary for commercial UAS operations

FAA DroneZone – <https://faadronezone.faa.gov/>

Drone registration, operator certificates, waiver management

FAA B4UFLY –

https://www.faa.gov/uas/recreational_flyers/where_can_i_fly/b4ufly

Pre-flight airspace check, TFRs, LAANC authorization

FAA Part 107 Waivers –

https://www.faa.gov/uas/commercial_operators/part_107_waivers

Waiver applications for operations beyond standard 107 rules

Aloft — Field Airspace Map – <https://www.aloft.ai/>

Mobile-friendly real-time airspace, LAANC, flight logging

AIR QUALITY & WILDFIRE SMOKE

[AirNow Fire & Smoke Map](https://www.fire.airnow.gov)

<https://www.fire.airnow.gov>

Current fire locations, smoke plumes, PM2.5 readings by location

[AirNow Interactive Map](https://gispub.epa.gov/airnow)

<https://gispub.epa.gov/airnow>

Real-time AQI monitors, hourly NowCast, 48-hr forecast overlays

[NOAA Smoke Forecast](https://airquality.weather.gov)

<https://airquality.weather.gov>

48-hr smoke concentration forecast, HRRR-Smoke model

